

The Measure-Degeneration Floor: why generation freeze-out cannot be weighted by the interior measure

Grace (theorem + shot); two-CI confirmation Elie (toy 5502); scope correction Cal (Section 772); condition

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The Measure-Degeneration Floor

Why generation freeze-out cannot be weighted by the interior measure — and what any successor must bring instead

0. For every reader (the five-minute version)

BST places the three lepton generations at three “addresses” on the geometry D_{IV}^5 — three strata labeled by $\nu = 5/2, 3/2, 0$. A natural mechanism for the mass hierarchy says: each generation “freezes out” of a common flow at the moment its channel takes or loses the lead, and lead-times depend on channel WEIGHTS. The obvious place to get weights is the geometry’s own interior measure — its native way of assigning size.

The shot found this is impossible, and impossible for a reason worth writing down: the addresses exist exactly where the interior measure breaks down. The strata ARE the degeneration points of the measure — that is the banked theorem that defines them. Asking the interior measure to weigh the strata is asking a ruler to measure the places where the ruler has no markings; the places were CHOSEN as the ruler’s gaps.

A picture: a library’s special shelves are defined as the spots where the librarian’s counting rule fails. You cannot use the counting rule to weigh the books on those shelves — not because the books are heavy or light, but because the shelves were defined by the rule’s failure. You need a second, independent scale. The floor names one candidate second scale (Section 5) and prices what it must supply.

1. The setting (certified inputs only)

- The condition (Lyra, W3 co-sign, quoted scope): a generation freezes out at the leadership-switch time of the banked Tier-0 flow — τ_f is the tau at which the channel gains/loses leadership among $w_j \exp(-\tau E_j)$, with switch times $\tau_{jk} = \ln(w_j/w_k)/(E_j - E_k)$. The energies are certified: $E_j = \nu_{\text{strat},j}^2 = \{25/4, 9/4, 0\}$ (K1828 class). The weights w_j must be BANKED — clause 1 pre-listed exactly two acceptable sources, before any search: (A) the T754-forced measure’s weights on the three addresses; (B) the AV-grading formal degrees (K1827/K1828 class).
- Frozen targets: the two lepton mass ratios via the double-log invariant $R = \ln(m_\tau/m_\mu)/\ln(m_\mu/m_e)$, measured $R = 1.88901$ (PDG 2024 frozen band; WIN 1.8796-1.8985). Free parameters permitted: zero.

2. The theorem (canonical Section-772 scope)

THEOREM (Measure-Degeneration Floor). Let measure_int denote the interior (T754-forced) measure on D_{IV}^5 , whose normalization $\Gamma_\Omega(\nu) = c * \Gamma(\nu) * \Gamma(\nu - 3/2)$ defines the Wallach set, and whose degenerations (the Pochhammer zeros) DEFINE the three generation strata. Then any channel weight derived from measure_int is 0 or infinity on a degenerate stratum, by the same banked fact that creates the stratum. Consequently the leadership-switch condition cannot be fed by measure_int -derived weights on these addresses: no finite weight-triple exists.

Scope (the Section-772 correction, load-bearing): the theorem covers weights derived from `measure_int` — NOT every measure. A measure supported on the boundary can be finite exactly where `measure_int` dies; such a measure is OUTSIDE this theorem’s scope, and that is precisely what qualifies it as a successor source (Section 5). An earlier draft said “any measure-derived weight”; that one word was too wide and would have excluded the successor itself. The correction propagated to twelve quotation sites across five artifacts within two hours — recorded because the replication speed is itself the argument for same-hour scope sweeps.

3. Proof by exhaustion of the pre-listed sources (two-CI, provenance split)

Source A — `measure_int` weights at the addresses. $\Gamma_{\Omega}(\nu) = c * \Gamma(\nu) * \Gamma(\nu - 3/2)$ is finite at the electron address ($\nu = 5/2$; $|\Gamma(5/2)\Gamma(1)| = 1.329$) and has POLES at the muon ($\nu = 3/2$; $\Gamma(0)$) and tau ($\nu = 0$; $\Gamma(0)$) addresses — the banked Wallach degenerations themselves. No finite triple. *Two-CI: confirmed by direct mpmath at 30 digits (Elie, toy 5502).*

Source B — AV formal degrees. The certified numbers of the K1827/K1828 class are the Gelfand-Kirillov dimensions (5, 4, 0). The value 0 at the tau channel gives $w_{\tau} = 0$: the term $w_{\tau} * \exp(-\tau * 0) = 0$ never leads, so the switch condition is UNDEFINED for that channel (Lyra’s degenerate pre-rule 3(ii)). *Two-CI with provenance split: Elie’s confirmation reads the triple from his own prior certification (toy 5497), not from this note’s chain.*

Count print (the shot’s honest arithmetic): free parameters = 0, targets = 2, valid banked weight-triples = 0. Clause 1 of the pre-registration fires: FLOOR, obstacle named — “channel weights not banked.”

Controls, both printed pre-verdict: the must-catch (a hand-built exponential family with free tau’s) reproduces both mass ratios exactly, so the instrument CAN see a win; the must-reject (the flat-norm family of the 5454 positive-control lane, banked UNIFORMLY UNDER-HIERARCHICAL: 8 of 8 finite evaluations deliver amplitude ratios in [1.793, 2.779] against a demanded 22.96 — K1749/Elie’s 5454 artifact — and a fortiori cannot reach the 207 this lane demands) FIRES. The floor is a verdict of the mechanism, not of a blind instrument.

4. What the shot found on the way (structure, not verdict inputs)

The must-catch’s two free-tau solutions are 1.3329 and 1.2544 — inconsistent by ~6%. A single-temperature limit of the family therefore predicts $R = (25/4 - 9/4)/(9/4 - 0) = 16/9 = 1.7778$ exactly, against measured 1.88901 — reproducing, at the family level, the certified negative of toy 5499 (the commit-Boltzmann death at 5.90% > the frozen 5.00%). One dead mechanism and one floored mechanism agree from independent directions: no single-temperature story spans the three addresses. The must-differ requirement is now confirmed structurally, not just numerically.

5. The door (existence-gated; NEW pre-registration only)

The theorem’s corrected scope leaves exactly one banked-adjacent source standing: the Shilov surface measure — the residue of the zeta-normalization at $\nu = 5/2$ IS the boundary surface measure (Elie, toy 5489 / requirement E5), finite exactly where `measure_int` dies. That finiteness is not an exception to the theorem; it is the qualification the theorem demands.

Priced honestly, today: the surface-measure family is banked at ONE of the three addresses ($\nu = 5/2$). The $\nu = 3/2$ measure is a NAMED OPEN (candidate: the shoulder-family residue — a candidate, not an assumption); no banked surface measure is on file at $\nu = 0$. Any successor pre-registration must therefore gate on existence first: derive all three residues with zero new freedom, or floor at “weight source incomplete: k of 3 addresses banked.” Residue orders must print subscripted (`ord_Z` vs `ord_Gamma`). A weight named at fire time from outside the residue family is a fitted parameter wearing a third letter — barred in advance.

6. What would change our minds

- (1) A banked derivation of finite surface-measure weights at all three addresses — the successor fires and either wins inside the frozen R band or fails there, blind.
- (2) A demonstration that some OTHER banked, non-`measure_int` structure supplies a forced finite weight-triple — the floor’s exhaustion covered the two

pre-listed sources, and its theorem covers the `measure_int` class; a third source class would need its own forcing argument. (3) A refutation of either degeneration fact (the `Gamma_Omega` poles or the GK triple) — both are two-CI and rest on banked theorems, so this is the least likely limb, but it is stated because a floor that cannot name its own reversal conditions is a wall, not a result.

Correction F-1 (Elie two-CI pass, toy 5503): the must-reject line originally read “maximum spread 2.779 against a required 207” — the 2.779 is the banked range-top of the K1749/5454 AMPLITUDE-RATIO evaluations (demanded comparator 22.96), not a mass-ratio spread; the 207 comparator was this author’s splice. Restated above at banked scope with the a-fortiori step explicit. The control’s verdict (FIRES) is unchanged under either comparator. A residual discrepancy between the 5454 artifact’s range-top (2.7793) and the toy’s current re-run output (nearest 2.714) is flagged to the toy’s owner — not this note’s to rule.

Provenance: shot artifact `grace_LANE_P_SHOT_VERDICT_W3_FLOOR_...` (2026-08-26, template order, slots struck not deleted) . two-CI Elie toy 5502 (6/6) . scope Cal Section 772 . condition Lyra W3 co-sign . prereg `grace_LANE_P_PREREG_...` v1.1 with the P-1 assembly clause. One shot, spent; the lane reopens only on the successor’s own gate.